```
from operator import itemgetter

class PC:
     def __init__(self, id, net_name, price, OS_id):
          self.id = id
          self.net_name = net_name
          self.price = price
          self.OS_id = OS_id

class OS:
     def __init__(self, id, name):
          self.id = id
          self.name = name

class PCOS:
     def __init__(self, PC_id, OS_id):
          self.PC_id = PC_id
          self.OS_id = OS_id

Operating_Systems = [
     OS(1, 'Widowz OS'),
     OS(2, 'LuxMax OS'),
     OS(3, 'VkusnoITochka'),

     OS(11, 'Other Widowz OS'),
     OS(22, 'Other LuxMax OS'),
     OS(33, 'Other VkusnoITochka'),
]

Personal_Computers = [
     PC(1, 'Mercury', 57900, 1),
     PC(2, 'Venus', 108200, 2),
     PC(3, 'Earth', 149600, 3),
     PC(4, 'Mars', 227900, 3),
     PC(5, 'Jupiter', 778600, 3),
]

PC_OS = [
     PCOS (1, 1),
     PCOS (2, 2),
     PCOS (3, 3),
     PCOS (4, 3),
     PCOS (5, 3),

     PCOS (1, 11),
     PCOS (2, 22),
     PCOS (3, 33),
     PCOS (4, 33),
     PCOS (5, 33),
]
```

```python
def main():
    one_to_many = [(pc.net_name, pc.price, os.name)
    for pc in Personal_Computers
    for os in Operating_Systems
    if pc.OS_id == os.id]

    many_to_many_temp = [(os.name, pcos.PC_id, pcos.OS_id)
    for os in Operating_Systems
    for pcos in PC_OS
    if os.id == pcos.OS_id]

    many_to_many = [(pc.net_name, pc.price, OS_name)
    for OS_name, PC_id, OS_id in many_to_many_temp
    for pc in Personal_Computers if pc.id == PC_id]

    print("Задание 1")
    res_1 = []
    for i in one_to_many:
        if 'OS' in i[2]:
            res_1.append(i)
    print(res_1)

    print('Задание 2')
    res_2_unsorted = []
    for os in Operating_Systems:
        pc_price = [price for net_name, price, os_name in one_to_many if
os_name == os.name]
        if len(pc_price) > 0:
            avg_price = sum(pc_price)/len(pc_price)
            res_2_unsorted.append((os.name, round(avg_price, 2)))
    res_2 = sorted(res_2_unsorted, key = itemgetter(1))
    print(res_2)

    print('Задание 3')
    res_3 = []
    for i in many_to_many:
        for j in i[0]:
            if j[0] == 'M':
                res_3.append((i[0], i[2]))
    print(res_3)

if __name__ == '__main__':
    main()
```

Результаты выполнения программы:

```
Задание 1
[('Mercury', 57900, 'Widowz OS'), ('Venus', 108200,
'LuxMax OS')]
Задание 2
[('Widowz OS', 57900.0), ('LuxMax OS', 108200.0),
('VkusnoITochka', 385366.67)]
Задание 3
[('Mercury', 'Widowz OS'), ('Mars', 'VkusnoITochka'),
('Mercury', 'Other Widowz OS'), ('Mars', 'Other
VkusnoITochka')]
```